

Assignment to simulate basic probability experiments to apply conditional probability and total probability.

1. Simulate Conditional Probability (Card Drawing)

Problem Statement:

A card is drawn from a standard deck of 52 cards.

Let event A be "card is a King", and event B be "card is a face card (Jack, Queen, or King)".

Tasks:

- Simulate drawing a card.
- Estimate $P(A)$, $P(B)$, and $P(A|B)$ (i.e., probability that the card is a King given it's a face card).
- Compare with theoretical values.

2. Conditional Probability (Dice Rolling)

Problem Statement:

Two fair six-sided dice are rolled.

Let:

- A = Sum of dice is 8
- B = At least one die shows 3

Tasks:

- Simulate rolling two dice.
- Compute $P(A)$, $P(B)$, and $P(A|B)$
- Validate results with theoretical calculation.

3. Total Probability (Machine Fault Detection)

Problem Statement:

A factory has 3 machines A, B, and C.

- Machine A produces 30% of items, B produces 45%, and C produces 25%.
- The defect rates are: A - 2%, B - 3%, C - 4%

Let D be the event that a randomly selected item is defective.

Tasks:

- Simulate 10,000 items being produced.
- Estimate $P(D)$ using the law of total probability.
- Also estimate conditional probability $P(A|D)$, $P(B|D)$, $P(C|D)$ (Bayes' Theorem).